event_log_1.csv — column reference

339,452 rows · 2,412 cases · 5 toolchains · 176 circuits · 15 algorithms · run 2026-09-22 19:44 → 09-23 20:07

Row = one pass executed during one compilation. **Case** = one compilation (circuit × toolchain × repetition).

Case identity

Column	Values in this file	Meaning
`case_id`	2,412 unique, e.g. `qiskit__qft_indep_n008__r1`	toolchain + circuit + repetition. **PM case ID**
`run_token`	2,412 unique hex strings	unique run ID; bookkeeping only
`toolchain`	bqskit 290,375 · qiskit 26,082 · cirq 10,647 · kirin 8,946 · pennylane 3,402	compiler
`rep`	0, 1, 2 (~113k rows each)	repetition number
`status`	`ok` only	failed cases are excluded

Circuit

Column	Values in this file	Meaning
`circuit_id`	176 unique, e.g. `cdkm_ripple_carry_adder_indep_n004`	input circuit
`algorithm`	15: vqe_two_local, qaoa, qpeexact, qft, vqe_su2, cdkm_ripple_carry_adder, wstate, qnn, ghz, graphstate, qwalk, dj, randomcircuit, grover, shor	algorithm family
`num_qubits`	2–32 (2,3,4,5,6,7,8,10,12,14,16,18,20,24,32)	circuit width
`selection_tier`	`CORE` only	main comparison set
`portability_tier`	`A` 180,261 · `A-EXT` 159,191	A = strict QASM2; A-EXT = uses Qiskit gate extensions
`input_sha16`	176 unique	input fingerprint; identical across toolchains for a circuit
`topology`	`grid` only	target coupling map

Step (pass)

Column	Values in this file	Meaning
`activity`	78 unique pass names	pass name. **PM activity** (per toolchain)
`stage`	control 198,961 · partitioning 44,207 · synthesis 41,660 · optimization 22,035 · analysis 13,509 · cleanup 5,817 · setup 4,296 · layout 4,191 · routing 2,481 · ir_lowering 1,836 · scheduling 459	ontology stage. **PM activity** (cross-toolchain)
`event_index`	0–13,688	position in the case. **Sort by this**

`seq`	0–13,688 (same as `event_index`)	internal order counter
`phase`	`compile` only	compile phase (Kirin `load` events: none)
`activity_module`	59 unique, e.g. `bqskit.passes.util.unfold`	source module of the pass
`pass_kind`	container 198,961 · pass 91,414 · transformation 14,667 · analysis 11,415 · transformer 10,140 · internal 7,569 · transform 3,402 · configured 1,377 · router 507	toolchain's own pass category (see table below)
`parent_activity`	18 unique + empty (13,587); top: Workflow, IfThenElsePass, ForEachBlockPass, stage:optimization	enclosing pass; Qiskit: its native stage
`nesting_depth`	0–12	0 = top level
`is_container`	True 204,763 · False 134,689	pass that only/also runs other passes
`iteration`	1–5,828	nth run of this activity in the case

Effect

Column	Values in this file	Meaning
`start_unix_ns`	1790106241467773000 – 1790194045010920000	start time, ns since 1970. **PM timestamp**
`duration_ns`	0 – 532,580,144,042 (median 40.3 ms)	pass runtime, tracing overhead removed
`ir_changed`	True 282,125 · False 57,327	pass changed the circuit
`changed_method`	circuit_fingerprint (bqskit) · dag_fingerprint (qiskit) · circuit_equality (cirq) · rewrite_result (kirin) · tape_fingerprint (pennylane)	how `ir_changed` was measured
`size_before`	1 – 100,144 (median 9)	IR size before the pass
`size_after`	0 – 100,144 (median 15)	IR size after the pass
`mem_peak_bytes`	empty everywhere	memory reps are excluded; merge with `--include-mem-reps`

Per toolchain

	Qiskit	Cirq	PennyLane	Kirin	BQSKit
Cases / circuits	528 / 176	507 / 169	486 / 162	459 / 153	432 / 144
Rows per case	49	21	7	20	672
Distinct passes	28	11	6	11	22
`pass_kind` values	transformation, analysis	transformer, router	transform	configured, internal	pass, container
`nesting_depth` max	0	1	0	4	12
`iteration` max	10	3	2	6	5,828
`parent_activity`	`stage:*`	enclosing transformer	empty	enclosing pass	Workflow / IfThenElse / ForEachBlock
`size_*` unit	DAG ops	ops	tape ops	IR statements	ops
Changed % (leaf rows)	14.7	60.0	61.8	39.4	87.1
Median / max step	0.004 ms / 130 ms	1.2 ms / 15.4 s	1.2 ms / 532.6 s	1.2 ms / 195 ms	49.3 ms / 156 s

Notes

- BQSKit is 86% of rows: mine pass-level models **per toolchain**; average per case, not per row.
- Drop `stage == control` (pure BQSKit wrappers) before stage-level mining.
- Qiskit `analysis` passes never change IR by design; transformation-only changed rate is 26.2%.
- `ir_changed` definitions differ per toolchain; BQSKit's fingerprint includes gate parameters, so re-instantiation counts as a change.
- Longest single step: PennyLane `transpile` on `grover_indep_n006`, 532.6 s.